Benjamin Walton

• ben.walton226@proton.me • Lexington, KY, USA •

Project Experience

Automated Product Seasonality Clustering & Live Inventory Simulation (SQL & Power BI)

- Created an end-to-end supply chain analytics pipeline managing daily sales data for 50 SKUs across 5 national distribution centers
- Produced an interactive yearly seasonality report and daily inventory health interface tracking stockout risks and creating automated daily product reorder tables
- Resolved a key inventory logic flaw within a synthetic dataset and simulated new synthetic data, giving three distinct seasonal demand curves.
- Automated product classification into seasonal clusters by leveraging Scikit-learn K-Means with Min-Max normalized sales data.
- Validated model flexibility by creating random shifts to ensure classification accuracy

Real-Time Operational Dashboard: Visualizing Logistics and Flight Disruptions

- Constructed a simulated real-time flight operations pipeline processing historical flight data (2M+ records) to model live domestic flight disruptions.
- Engineered time-series data simulation scripts using Pandas to segment massive historical datasets into sequential, 10-minute micro-batches to replicate operations data streams.
- Built an interactive live-mapping application using Plotly Dash and custom callback logic to visualize cancellations as geographic hotspots.
- Developed dynamic multi-airline tracking metrics comparing cancellations across the big four domestic airlines (Southwest, American, Delta, United) to deliver insights within a live updating carrier performance dashboard

Skysolver (ongoing experiential project)

- Used Python, Numpy, and Pandas to streamline work tasks, and automate large scale file generation and organization within Windows.
- Aim to simplify complicated airline scheduling during high cancellations/ delays
- To simulate operations, built backend flight and crew systems from scratch
- Deep dive into planning and executing scalable data objects to solve real world problems

The Association Between New Housing in Large Metropolitan Areas and Migration to Areas with Higher Crime (2021)

- Engineered cross-application Excel automations to streamline workflows, eliminate manual entries, and optimize processes across external system applications.
- Aggregated and consolidated 50+ disparate datasets, executing advanced data-cleaning methodologies to establish a unified source for downstream analysis.
- Developed an econometric isolation model to decouple variables and isolate the statistical impact of new housing developments on demographic migration patterns.

Skills: SQL | Excel/ Google Sheets | Python (Numpy, Pandas, Scikit-learn) | Power BI

Other Skills: Basic spoken Spanish | Skilled with computers and troubleshooting

---

Education

University of Kentucky - *Aug 2017- May 2022*

- **Mathematical Economics**, Bach. of Science | **Finance**, Bach. of Business Administration
  Minor in Philosophy

Relevant Coursework:

| Finance and accounting | Statistics and probability, including courses at the graduate level |
| Economics and econometrics | Multiple finance classes focusing on proficiency in Excel |

Current Employment

Darae & Friends Catering - Customer Experience Supervisor (2022 - Present)

- Managed reorders, spatial warehousing constraints, and optimized production workflows
- Developed my ability to hone-in on stakeholder cues to manage, predict and meet needs
- Effectively trained new staff in a friendly manner, while ensuring work standards were clear